
Discrete Mathematics

Complete Answer Key

Practices 1–8

*This document provides correct answers and brief solution sketches for **every** exercise across all eight practice sets. Final numerical answers are highlighted with $\boxed{\cdot}$. For detailed worked solutions with pedagogical commentary, see the individual solution documents.*

Contents

Practice 1: Set Theory

Problem 1

What does $\{\emptyset\}$ represent and how many elements does it have?

$\{\emptyset\}$ is the set whose only element is the empty set. It has $\boxed{1}$ element.

Problem 2

Which statements are true?

- 2.1. $\emptyset \in \{\emptyset, \{\emptyset\}\}$ — **TRUE**. The set lists $\emptyset$ as an element.
- 2.2. $\{\emptyset\} \in \{\{\emptyset\}\}$ — **TRUE**. The sole element of $\{\{\emptyset\}\}$ is $\{\emptyset\}$.
- 2.3. $\emptyset \in \{\{\emptyset\}\}$ — **FALSE**. The sole element is $\{\emptyset\} \neq \emptyset$.

Problem 3

How many elements does $\{\{\emptyset\}\}$ have?

$\boxed{1}$. Its only element is $\{\emptyset\}$.

Problem 4

How many elements do the following sets have?

- 4.1. $\{1, 2, 1\}$: $\boxed{2}$ (duplicates are ignored; the set is $\{1, 2\}$).
- 4.2. $\{1, 2, \{1, 2\}\}$: $\boxed{3}$ (elements: 1, 2, $\{1, 2\}$).
- 4.3. $\{\{2\}\}$: $\boxed{1}$ (element: $\{2\}$).
- 4.4. $\{1, \{1\}\}$: $\boxed{2}$ (elements: 1 and $\{1\}$; these are distinct).
- 4.5. $\{1, \emptyset\}$: $\boxed{2}$ (elements: 1 and $\emptyset$).
- 4.6. $\{\{\emptyset\}, \emptyset\}$: $\boxed{2}$ (elements: $\{\emptyset\}$ and $\emptyset$).
- 4.7. $\{\{\emptyset\}, \{\emptyset\}\}$: $\boxed{1}$ (duplicate; the set is $\{\{\emptyset\}\}$).
- 4.8. $\{x \mid 2x = 1\}$ where $x \in \mathbb{R}$: $\boxed{1}$ (the set is $\{\frac{1}{2}\}$).

Problem 5

Is $\{1, 2\} \in \{1, 2, \{1, 3\}, \{1, 2, 3\}\}$?

NO. The elements of the right-hand set are 1, 2, $\{1, 3\}$, and $\{1, 2, 3\}$. None of these equals $\{1, 2\}$.

Problem 6

Give examples of sets that are elements of themselves.

In standard ZFC set theory (with the Axiom of Regularity), no set is an element of itself. However, in naive set theory or theories without regularity, the “set of all sets” $V = \{x \mid x = x\}$ would satisfy $V \in V$. Another classic example: if we define S as “the set of all sets that are not sets of numbers,” then $S \in S$ (Russell-type construction). In ZFC, **no such set exists**.

Problem 7

Give examples of A, B, C with $A \in B$, $B \in C$, but $A \notin C$.

Let $A = 1$, $B = \{1, 2\}$, $C = \{\{1, 2\}, 3\}$. Then $1 \in \{1, 2\}$, $\{1, 2\} \in \{\{1, 2\}, 3\}$, but $1 \notin \{\{1, 2\}, 3\}$ since the elements of C are $\{1, 2\}$ and 3 .

Problem 8

Prove: $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\} \iff a = c \text{ and } b = d$.

Proof. $(\Leftarrow)$ Trivial by substitution. $(\Rightarrow)$ From the equality, $\{a\}$ must equal either $\{c\}$ or $\{c, d\}$. If $\{a\} = \{c, d\}$ then $c = d = a$, and then $\{a, b\} = \{c\} = \{a\}$ forces $b = a = c = d$. If $\{a\} = \{c\}$ then $a = c$, and the remaining element gives $\{a, b\} = \{c, d\} = \{a, d\}$, so $b = d$. $\square$

Problem 9

Prove three set equalities/inclusions.

(a) $\{x \in \mathbb{Z} \mid \exists y, x = 6y\} = \{x \in \mathbb{Z} \mid \exists u, v, x = 2u \text{ and } x = 3v\}$.

Proof. $(\subseteq)$ If $x = 6y$ then $x = 2(3y)$ and $x = 3(2y)$. $(\supseteq)$ If $x = 2u$ and $x = 3v$, then x is divisible by both 2 and 3, hence by $\text{lcm}(2, 3) = 6$. $\square$

(b) $\{x \in \mathbb{R} \mid \exists y, x = y^2\} = \{x \in \mathbb{R} \mid x \geq 0\}$.

Proof. $(\subseteq)$ $x = y^2 \geq 0$. $(\supseteq)$ If $x \geq 0$, take $y = \sqrt{x}$, then $x = y^2$. $\square$

(c) $\{x \in \mathbb{Z} \mid \exists y, x = 6y\} \subset \{x \in \mathbb{Z} \mid \exists y, x = 2y\}$.

Proof. If $x = 6y$ then $x = 2(3y)$, so $\subseteq$ holds. The inclusion is strict since $2 = 2 \cdot 1$ is in the right set but 2 is not a multiple of 6. $\square$

Problem 10

Prove transitivity properties of $\subseteq$ and $\subset$.

(a) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proof. Let $x \in A$. Then $x \in B$ (since $A \subseteq B$), then $x \in C$ (since $B \subseteq C$). $\square$

(b) If $A \subseteq B$ and $B \subset C$, then $A \subset C$.

Proof. By (a), $A \subseteq C$. Since $B \subset C$, there exists $z \in C \setminus B$. If $A = C$, then $z \in A \subseteq B$, contradiction. So $A \neq C$, giving $A \subset C$. $\square$

(c) If $A \subset B$ and $B \subseteq C$, then $A \subset C$.

Proof. By (a), $A \subseteq C$. Since $A \subset B$, there exists $z \in B \setminus A$, and $z \in C$ since $B \subseteq C$. So $z \in C \setminus A$, hence $A \neq C$. $\square$

(d) If $A \subset B$ and $B \subset C$, then $A \subset C$.

Proof. Follows from (c) since $B \subset C$ implies $B \subseteq C$. □

Problem 11

Give examples of A, B, C, D, E with $A \subseteq B, B \in C, C \subseteq D, D \subseteq E$.

$A = \{1\}, B = \{1, 2\}, C = \{\{1, 2\}, 3\}, D = \{\{1, 2\}, 3, 4\}, E = \{\{1, 2\}, 3, 4, 5\}$.

Problem 12

Which statements are true for all sets A, B, C ?

- (a) If $A \notin B$ and $B \notin C$, then $A \notin C$. — **FALSE**. Counterexample: $A = 1, B = \{2\}, C = \{1\}$. Then $1 \notin \{2\}$ and $\{2\} \notin \{1\}$, but $1 \in \{1\}$.
- (b) If $A \neq B$ and $B \neq C$, then $A \neq C$. — **FALSE**. Counterexample: $A = C = \{1\}, B = \{2\}$.
- (c) If $A \in B$ and $B \subseteq C$, then $A \in C$. — **TRUE**. Since $A \in B$ and every element of B is in C , we get $A \in C$.
- (d) If $A \subseteq B$ and $B \subseteq C$, then $C \subseteq A$? — **FALSE**. Counterexample: $A = \emptyset, B = C = \{1\}$.
- (e) If $A \subseteq B$ and $B \in C$, does $A \notin C$? — **FALSE** (not necessarily). Counterexample: $A = \{1\}, B = \{1, 2\}, C = \{\{1\}, \{1, 2\}\}$. Then $A \subseteq B, B \in C$, and also $A \in C$.

Problem 13

Give examples of sets where every element is a subset of the set (transitive sets).

$\emptyset$ (vacuously true). Also $\{\emptyset\}$: its only element $\emptyset \subseteq \{\emptyset\}$. Also $\{\emptyset, \{\emptyset\}\}$: $\emptyset \subseteq \{\emptyset, \{\emptyset\}\}$ and $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$. In general, any ordinal number is a transitive set.

Problem 14

List the elements of 2^A .

14.1. $A = \emptyset$: $2^A = \{\emptyset\}$ (1 element).

14.2. $A = \{x\}$: $2^A = \{\emptyset, \{x\}\}$ (2 elements).

14.3. $A = \{1, a\}$: $2^A = \{\emptyset, \{1\}, \{a\}, \{1, a\}\}$ (4 elements).

14.4. $A = \{\emptyset, \{1\}, \{2\}\}$: 2^A has $2^3 = 8$ elements:

$$2^A = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \{\{2\}\}, \\ \{\emptyset, \{1\}\}, \{\emptyset, \{2\}\}, \{\{1\}, \{2\}\}, \{\emptyset, \{1\}, \{2\}\}\}$$

14.5. $A = \{1, \{3\}, \{1, 2\}\}$: 2^A has $2^3 = 8$ elements:

$$2^A = \{\emptyset, \{1\}, \{\{3\}\}, \{\{1, 2\}\}, \\ \{1, \{3\}\}, \{1, \{1, 2\}\}, \{\{3\}, \{1, 2\}\}, \{1, \{3\}, \{1, 2\}\}\}$$

Problem 15

Prove $\emptyset \subseteq A \cap B \subseteq A \cup B$.

Proof. $\emptyset \subseteq S$ for any set S (vacuous truth). If $x \in A \cap B$, then $x \in A$, so $x \in A \cup B$. Hence $A \cap B \subseteq A \cup B$. $\square$

Problem 16

Describe $\overline{A}$, $\overline{A \cup B}$, $\overline{C}$, $A \setminus C$, $C \setminus (A \cup B)$ with $U = \mathbb{Z}$, $A = \{2k \mid k \in \mathbb{Z}^+\}$ (positive evens), $B = \{2k - 1 \mid k \in \mathbb{Z}^+\}$ (positive odds), $C = \{x \in \mathbb{Z} \mid x < 10\}$.

Here $A \cap B = \emptyset$ and $A \cup B = \mathbb{Z}^+$ (every positive integer is even or odd).

- $\overline{A} = \{x \in \mathbb{Z} \mid x \leq 0 \text{ or } x \text{ is odd}\}$
- $\overline{A \cup B} = \{x \in \mathbb{Z} \mid x \leq 0\} = \{0, -1, -2, \dots\}$
- $\overline{C} = \{x \in \mathbb{Z} \mid x \geq 10\}$
- $A \setminus C = \{10, 12, 14, \dots\}$ (positive evens ≥ 10)
- $C \setminus (A \cup B) = \{x \in \mathbb{Z} \mid x \leq 0\} = \{\dots, -2, -1, 0\}$

Problem 17

A = positive evens, B = positive odds, C = positive multiples of 3.

(a)

- $A \cap C = \{6, 12, 18, \dots\}$ = positive multiples of 6.
- $B \cup C = \{n \in \mathbb{Z}^+ \mid n \text{ is odd or } 3 \mid n\} = \{1, 3, 5, 6, 7, 9, 11, 12, 13, \dots\}$.
- $B \setminus C = \{1, 5, 7, 11, 13, 17, \dots\}$ = positive odd integers not divisible by 3.

(b) Yes, the distributive law $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ holds for all sets. Since $A \cap B = \emptyset$ (no positive integer is both even and odd), we get $A \cap (B \cup C) = \emptyset \cup (A \cap C) = A \cap C$ = positive multiples of 6. Independently, $A \cap (B \cup C)$ consists of positive even integers that are odd or divisible by 3; the “odd” part is impossible, so this is positive evens divisible by 3, i.e., multiples of 6. Both sides agree. $\checkmark$

Problem 18

For any set A :

$$\boxed{A \cap \emptyset = \emptyset}, \quad \boxed{A \cup \emptyset = A}, \quad \boxed{A \setminus \emptyset = A}, \quad \boxed{A \setminus A = \emptyset}, \quad \boxed{\emptyset \setminus A = \emptyset}.$$

Problem 19

Compute:

- $\emptyset \setminus \{\emptyset\} = \boxed{\emptyset}$ (nothing to remove from the empty set).

- $\{\emptyset\} \setminus \{\emptyset\} = \boxed{\emptyset}$ (the sole element $\emptyset$ is removed).
- $\{\emptyset, \{\emptyset\}\} \setminus \emptyset = \boxed{\{\emptyset, \{\emptyset\}\}}$ (nothing is removed).
- $\{\emptyset, \{\emptyset\}\} \setminus \{\emptyset\} = \boxed{\{\{\emptyset\}\}}$ (the element $\emptyset$ is removed; $\{\emptyset\}$ remains).

Problem 20

Prove equivalent characterizations.

Part 1: $A \subseteq B \iff \overline{B} \subseteq \overline{A} \iff A \cup B = B \iff A \cap B = A$.

Proof sketch. Close the cycle $A \subseteq B \Rightarrow A \cup B = B \Rightarrow A \cap B = A \Rightarrow \overline{B} \subseteq \overline{A} \Rightarrow A \subseteq B$.
 $(A \subseteq B) \Rightarrow (A \cup B = B)$: every element of A is already in B , so $A \cup B = B$.
 $(A \cup B = B) \Rightarrow (A \cap B = A)$: if $x \in A$ then $x \in A \cup B = B$, so $x \in A \cap B$.
 $(A \cap B = A) \Rightarrow (\overline{B} \subseteq \overline{A})$: $A \cap B = A$ gives $A \subseteq B$; if $x \notin B$ then $x \notin A$, i.e. $\overline{B} \subseteq \overline{A}$.
 $(\overline{B} \subseteq \overline{A}) \Rightarrow (A \subseteq B)$: contrapositive — if $x \in A$ then $x \notin \overline{A}$, so $x \notin \overline{B}$, i.e. $x \in B$. $\square$

Part 2: $A \cap B = \emptyset \iff A \subseteq \overline{B} \iff B \subseteq \overline{A}$.

Proof sketch. Disjointness means every element of A is outside B (i.e. in $\overline{B}$), and symmetrically every element of B is outside A . Cyclic implications close the loop. $\square$

Part 3: $A \cup B = U \iff \overline{A} \subseteq B \iff \overline{B} \subseteq A$.

Proof sketch. If $A \cup B = U$ and $x \notin A$, then $x \in B$; so $\overline{A} \subseteq B$. If $\overline{A} \subseteq B$ and $x \notin B$, then $x \notin \overline{A}$, i.e. $x \in A$; so $\overline{B} \subseteq A$. If $\overline{B} \subseteq A$ then for any $x \in U$, either $x \in B$ or $x \in \overline{B} \subseteq A$, so $A \cup B = U$. $\square$

Problem 21

Prove $(A \setminus B) \setminus C = A \setminus (B \cup C)$. Is $C \subseteq A$ equivalent?

Proof. $x \in (A \setminus B) \setminus C \iff x \in A \wedge x \notin B \wedge x \notin C \iff x \in A \wedge x \notin (B \cup C) \iff x \in A \setminus (B \cup C)$. $\square$

The condition $C \subseteq A$ is **not** equivalent to this identity. The identity $(A \setminus B) \setminus C = A \setminus (B \cup C)$ holds for *all* sets A, B, C , regardless of whether $C \subseteq A$.

Problem 22

Prove $(A \setminus B) \setminus C = (A \setminus C) \setminus (B \setminus C)$.

Proof. $x \in (A \setminus C) \setminus (B \setminus C) \iff x \in A \wedge x \notin C \wedge \neg(x \in B \wedge x \notin C)$. Since $x \notin C$, the condition $\neg(x \in B \wedge x \notin C)$ reduces to $x \notin B$. So this equals $x \in A \wedge x \notin B \wedge x \notin C = x \in (A \setminus B) \setminus C$. $\square$

Problem 23

Prove $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Proof. Base: $n = 1$: $1 = \frac{1 \cdot 2}{2}$. Inductive step: assume true for $n = k$. Then $\sum_{i=1}^{k+1} i = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$. $\square$

Problem 24

Prove $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Proof. Base: $n = 1$: $1 = \frac{1 \cdot 2 \cdot 3}{6}$. Inductive step: assume for k . Then $\frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)[k(2k+1)+6(k+1)]}{6} = \frac{(k+1)(2k^2+7k+6)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$. $\square$

Problem 25

Prove $1 + 4 + 7 + \cdots + (3n - 2) = \frac{n(3n-1)}{2}$.

Proof. Base: $n = 1$: $1 = \frac{1 \cdot 2}{2}$. Inductive step: $\frac{k(3k-1)}{2} + (3(k+1) - 2) = \frac{k(3k-1)+2(3k+1)}{2} = \frac{3k^2+5k+2}{2} = \frac{(k+1)(3k+2)}{2} = \frac{(k+1)(3(k+1)-1)}{2}$. $\square$

Problem 26

Prove $\sum_{i=1}^n \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1}$.

Proof. Base: $n = 1$: $\frac{1}{1 \cdot 3} = \frac{1}{3}$. Inductive step: $\frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} = \frac{k(2k+3)+1}{(2k+1)(2k+3)} = \frac{2k^2+3k+1}{(2k+1)(2k+3)} = \frac{(k+1)(2k+1)}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$. $\square$

Problem 27

Prove $1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$.

Proof. Base: $n = 1$: $1 = 2^1 - 1$. Inductive step: $(2^k - 1) + 2^k = 2 \cdot 2^k - 1 = 2^{k+1} - 1$. $\square$

Problem 28

Prove $1 + r + r^2 + \cdots + r^{n-1} = \frac{1-r^n}{1-r}$ for $r \neq 1$.

Proof. Base: $n = 1$: $1 = \frac{1-r}{1-r}$. Inductive step: $\frac{1-r^k}{1-r} + r^k = \frac{1-r^k+r^k(1-r)}{1-r} = \frac{1-r^{k+1}}{1-r}$. $\square$

Problem 29

Prove $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

Proof. Base: $n = 1$: $1 = 1^2$. Inductive step: $k^2 + (2(k+1) - 1) = k^2 + 2k + 1 = (k+1)^2$. $\square$

Problem 30

Prove $2 + 4 + 6 + \cdots + 2n = n^2 + n$.

Proof. Base: $n = 1$: $2 = 1 + 1$. Inductive step: $(k^2 + k) + 2(k+1) = k^2 + 3k + 2 = (k+1)^2 + (k+1)$. $\square$

Problem 31

Prove by induction: $a^{m+n} = a^m \cdot a^n$.

Proof. Fix m ; induct on n . Base: $n = 0$: $a^{m+0} = a^m = a^m \cdot 1 = a^m \cdot a^0$. Inductive step: $a^{m+(k+1)} = a^{(m+k)+1} = a^{m+k} \cdot a = (a^m \cdot a^k) \cdot a = a^m \cdot a^{k+1}$. $\square$

Problem 32

Prove by induction: $(ab)^k = a^k b^k$.

Proof. Base: $k = 0$: $(ab)^0 = 1 = a^0 b^0$. (Or $k = 1$: $(ab)^1 = ab = a^1 b^1$.) Inductive step: $(ab)^{k+1} = (ab)^k \cdot ab = a^k b^k \cdot ab = a^{k+1} b^{k+1}$ (using commutativity of multiplication). $\square$

Problem 33

Prove $n^2 > 2n + 1$ for $n \geq 3$.

Proof. Base: $n = 3$: $9 > 7$. Inductive step: assume $k^2 > 2k + 1$ for some $k \geq 3$. $(k + 1)^2 = k^2 + 2k + 1 > (2k + 1) + 2k + 1 = 4k + 2 \geq 2(k + 1) + 1 + (2k - 1) > 2(k + 1) + 1$ since $2k - 1 \geq 5 > 0$. $\square$

Problem 34

Prove $2^n > n^2$ for $n \geq 5$.

Proof. Base: $n = 5$: $32 > 25$. Inductive step: assume $2^k > k^2$ for $k \geq 5$. $2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k + 1)^2 = k^2 + 2k + 1$, i.e., $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$. $\square$

Problem 35

Prove $a^3 > 3a^2 + 3a + 1$ for $a \geq 4$.

Proof. Base: $a = 4$: $64 > 48 + 12 + 1 = 61$. $\checkmark$

Inductive step: assume $k^3 > 3k^2 + 3k + 1$ for $k \geq 4$. We show $(k + 1)^3 > 3(k + 1)^2 + 3(k + 1) + 1$. $(k + 1)^3 = k^3 + 3k^2 + 3k + 1 > (3k^2 + 3k + 1) + 3k^2 + 3k + 1 = 6k^2 + 6k + 2$. We need $6k^2 + 6k + 2 \geq 3k^2 + 9k + 7$, i.e., $3k^2 - 3k - 5 \geq 0$, which holds for $k \geq 2$. $\square$

Problem 36

Prove $5^{2n} + 3 \cdot 2^{5n-2}$ is divisible by 7 for all positive integers n .

Proof. Base: $n = 1$: $25 + 3 \cdot 8 = 25 + 24 = 49 = 7 \cdot 7$. $\checkmark$

Inductive step: assume $7 \mid (5^{2k} + 3 \cdot 2^{5k-2})$. Then $5^{2(k+1)} + 3 \cdot 2^{5(k+1)-2} = 25 \cdot 5^{2k} + 32 \cdot 3 \cdot 2^{5k-2} = 25 \cdot 5^{2k} + 25 \cdot 3 \cdot 2^{5k-2} + 7 \cdot 3 \cdot 2^{5k-2} = 25(5^{2k} + 3 \cdot 2^{5k-2}) + 7 \cdot 3 \cdot 2^{5k-2}$. Both terms are divisible by 7. $\square$

Problem 37

Prove $3^{n+2} + 4^{2n+1}$ is divisible by 13 for all $n \geq 0$.

Proof. Base: $n = 0$: $9 + 4 = 13$. $\checkmark$ (Or $n = 1$: $27 + 64 = 91 = 7 \cdot 13$.)

Inductive step: assume $13 \mid (3^{k+2} + 4^{2k+1})$. Then $3^{k+3} + 4^{2k+3} = 3 \cdot 3^{k+2} + 16 \cdot 4^{2k+1} = 3(3^{k+2} + 4^{2k+1}) + 13 \cdot 4^{2k+1}$. Both terms are divisible by 13. $\square$

Problem 38

Prove $5^{2n} + 7$ is divisible by 8 for all positive integers n .

Proof. $5^{2n} = 25^n \equiv 1^n = 1 \pmod{8}$, so $5^{2n} + 7 \equiv 1 + 7 = 8 \equiv 0 \pmod{8}$.

Alternatively by induction: Base: $n = 1$: $25 + 7 = 32 = 8 \cdot 4$. Inductive step: $5^{2(k+1)} + 7 = 25 \cdot 5^{2k} + 7 = 25(5^{2k} + 7) - 25 \cdot 7 + 7 = 25(5^{2k} + 7) - 168$. Both $25(5^{2k} + 7)$ and $168 = 8 \cdot 21$ are divisible by 8. $\square$

Problem 39

Prove $3^{3n+1} + 2^{n+1}$ is divisible by 5 for $n \in \mathbb{N}$.

Proof. Base: $n = 0$: $3 + 2 = 5$. $\checkmark$

Inductive step: $3^{3(k+1)+1} + 2^{k+2} = 27 \cdot 3^{3k+1} + 2 \cdot 2^{k+1} = 27 \cdot 3^{3k+1} + 27 \cdot 2^{k+1} - 25 \cdot 2^{k+1} = 27(3^{3k+1} + 2^{k+1}) - 25 \cdot 2^{k+1}$. By hypothesis $5 \mid 27(3^{3k+1} + 2^{k+1})$ (since 5 divides the inner expression) and $5 \mid 25 \cdot 2^{k+1}$. $\square$

Problem 40

Prove $9^n - 1$ is divisible by 8 for $n \in \mathbb{N}$.

Proof. $9^n - 1 = (9 - 1)(9^{n-1} + 9^{n-2} + \cdots + 1) = 8 \cdot \sum_{i=0}^{n-1} 9^i$.

Alternatively: $9 \equiv 1 \pmod{8}$, so $9^n \equiv 1 \pmod{8}$. $\square$

Problem 41

Prove $10^n - 1$ is divisible by 9 for $n \in \mathbb{N}$.

Proof. $10 \equiv 1 \pmod{9}$, so $10^n \equiv 1 \pmod{9}$, hence $9 \mid (10^n - 1)$.

Alternatively: $10^n - 1 = \underbrace{99 \dots 9}_n = 9 \cdot \underbrace{11 \dots 1}_n$. $\square$

Problem 42

Prove $n^2 - 1$ is divisible by 4 for all odd natural numbers.

Proof. Let $n = 2m + 1$ for $m \geq 0$. Then $n^2 - 1 = (2m + 1)^2 - 1 = 4m^2 + 4m = 4m(m + 1)$, which is divisible by 4. $\square$

(No induction needed; a direct proof suffices.)

Problem 43

Find and prove: $3 + 7 + 11 + \cdots + (4n - 1)$.

The sum is $\boxed{n(2n + 1)}$.

Proof. Base: $n = 1$: $3 = 1 \cdot 3$. $\checkmark$

Inductive step: $k(2k + 1) + (4(k + 1) - 1) = 2k^2 + k + 4k + 3 = 2k^2 + 5k + 3 = (k + 1)(2k + 3) = (k + 1)(2(k + 1) + 1)$. $\square$

Problem 44

Find and prove: $1 + 5 + 9 + \cdots + (4n + 1)$.

The last term $4n + 1$ is the $(n + 1)$ -th term of $1, 5, 9, \dots$, so the sum has $n + 1$ terms and equals $\boxed{(n + 1)(2n + 1)}$.

Proof. $\sum_{k=1}^{n+1} (4k - 3) = 4 \cdot \frac{(n + 1)(n + 2)}{2} - 3(n + 1) = (n + 1)[2(n + 2) - 3] = (n + 1)(2n + 1)$.

Check: $n = 1$: $1 + 5 = 6 = 2 \cdot 3$; $n = 2$: $1 + 5 + 9 = 15 = 3 \cdot 5$. $\checkmark$ $\square$

Problem 45

Find and prove: $-1 + 2 - 3 + 4 - \cdots - (2n - 1) + 2n$.

The sum is $\boxed{n}$.

Proof. Group in n pairs: $(-(2k - 1) + 2k) = 1$ for $k = 1, \dots, n$. Hence the total is $n \cdot 1 = n$.

Check: $-1 + 2 = 1$; $-1 + 2 - 3 + 4 = 2$; $-1 + 2 - 3 + 4 - 5 + 6 = 3$. ✓

□

Problem 46

Find and prove: $1 - 3 + 5 - 7 + \cdots + (4n - 3) - (4n - 1)$.

The sum is $\boxed{-2n}$.

Proof. Group in n pairs: $((4k - 3) - (4k - 1)) = -2$ for $k = 1, \dots, n$. Hence the total is $-2n$. Check: $1 - 3 = -2$; $1 - 3 + 5 - 7 = -4$; $1 - 3 + 5 - 7 + 9 - 11 = -6$. ✓

□

Problem 47

Prove $2^n > n^3$ and $n! > 100^n$ for large n .

(a) $2^n > n^3$ for $n \geq 10$.

Proof. Base: $n = 10$: $1024 > 1000$. Inductive step: assume $2^k > k^3$ for $k \geq 10$. Then $2^{k+1} = 2 \cdot 2^k > 2k^3$. We need $2k^3 \geq (k+1)^3$, i.e., $(1 + 1/k)^3 \leq 2$. For $k \geq 10$: $(1 + 1/k)^3 \leq (1.1)^3 = 1.331 < 2$. ✓

□

(b) $n! > 100^n$ for $n \geq 200$ (for instance).

Proof. Once $k \geq 200$, we have $k+1 > 100$, so $(k+1)! = (k+1) \cdot k! > 100 \cdot 100^k = 100^{k+1}$. The base case $200! > 100^{200}$ can be verified using Stirling's approximation. ✓

□

Problem 48

Tower of Hanoi: prove n disks can be moved in $2^n - 1$ moves.

Proof. Base: $n = 1$: move the single disk directly (1 move = $2^1 - 1$). Inductive step: to move $k+1$ disks from peg A to peg C , first move the top k disks from A to B ($2^k - 1$ moves by hypothesis), then move the largest disk from A to C (1 move), then move k disks from B to C ($2^k - 1$ moves). Total: $2(2^k - 1) + 1 = 2^{k+1} - 1$. ✓

□

Problem 49

Bernoulli's inequality: $(1+h)^n \geq 1+nh$ for $h > -1$, $n \in \mathbb{N}$.

Proof. Base: $n = 1$: $1+h \geq 1+h$. ✓

Inductive step: $(1+h)^{k+1} = (1+h)^k(1+h) \geq (1+kh)(1+h) = 1+(k+1)h+kh^2 \geq 1+(k+1)h$ since $kh^2 \geq 0$. (We used $1+h > 0$ to preserve the inequality direction when multiplying.) ✓

□

Problem 50

Prove Theorem 1 (basic set algebra laws) using membership relation.

“Theorem 1” typically refers to the fundamental identities:

1. Commutativity: $A \cup B = B \cup A$, $A \cap B = B \cap A$.
2. Associativity: $(A \cup B) \cup C = A \cup (B \cup C)$, similarly for $\cap$.
3. Distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
4. Identity: $A \cup \emptyset = A$, $A \cap U = A$.
5. Complement: $A \cup \overline{A} = U$, $A \cap \overline{A} = \emptyset$.

6. Idempotency: $A \cup A = A$, $A \cap A = A$.

7. De Morgan: $\overline{A \cup B} = \overline{A} \cap \overline{B}$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

Proof sketch. Each identity is proved by showing $x \in \text{LHS} \iff x \in \text{RHS}$ for arbitrary x , translating set operations to logical connectives ($\cup \rightarrow \vee$, $\cap \rightarrow \wedge$, $\bar{\cdot} \rightarrow \neg$) and applying standard propositional logic laws. $\square$

Problem 51

Prove Theorem 2 using membership relation.

“Theorem 2” typically includes additional identities such as: absorption ($A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$), involution ($\overline{\overline{A}} = A$), domination ($A \cup U = U$, $A \cap \emptyset = \emptyset$), and $\overline{\emptyset} = U$.

Proof sketch. As in Problem 50, each is proved via element-chasing, reducing to tautologies of propositional logic. For example, absorption: $x \in A \cup (A \cap B) \iff x \in A \vee (x \in A \wedge x \in B) \iff x \in A$. $\square$

Problem 52

Prove Theorem 2 using Theorem 1 identities as axioms (algebraic proofs, no element-chasing).

Proof sketch. For example, absorption: $A \cup (A \cap B) = (A \cap U) \cup (A \cap B) = A \cap (U \cup B) = A \cap U = A$, using identity, distributivity, domination, and identity laws from Theorem 1. Each Theorem 2 identity can be derived purely algebraically from Theorem 1. $\square$

Problem 53

Using Theorems 1 & 2, prove identities.

(1) $(A \cap B \cap X) \cup (A \cap B \cap C \cap X \cap Y) \cup (A \cap X \cap \overline{A}) = A \cap B \cap X$.

Proof sketch. The third term contains $A \cap \overline{A} = \emptyset$, so it vanishes. The second term is a subset of the first (it has extra constraints C and Y), so first $\cup$ second = first, by absorption. Result: $A \cap B \cap X$. $\square$

(2) $(A \cap B \cap C) \cup (\overline{A} \cap B \cap C) \cup \overline{B} \cup \overline{C} = U$.

Proof sketch. $(A \cap B \cap C) \cup (\overline{A} \cap B \cap C) = (A \cup \overline{A}) \cap B \cap C = B \cap C$, so the left side is $(B \cap C) \cup \overline{B} \cup \overline{C} = (B \cap C) \cup \overline{B \cap C} = U$ (De Morgan). $\square$

(3) $(A \cap B \cap C \cap \overline{X}) \cup (\overline{A} \cap C) \cup (\overline{B} \cap C) \cup (C \cap X) = C$.

Proof sketch. Factor out C : the left side is $C \cap [(A \cap B \cap \overline{X}) \cup \overline{A} \cup \overline{B} \cup X]$. With $S = A \cap B \cap \overline{X}$, note $\overline{A} \cup \overline{B} \cup X = \overline{S}$, so the bracket is $S \cup \overline{S} = U$. Hence the result is $C \cap U = C$. $\square$

Problem 54

Prove the set identities (all true; each by element-chasing, translating $\setminus$ as $\cap(\overline{\cdot})$ and $A \triangle B$ as “ x in exactly one of A, B ”).

(1) $A \cap (B \setminus C) = (A \cap B) \setminus C$.

$x \in A \wedge x \in B \wedge x \notin C$ on both sides. $\checkmark$

(2) $A \setminus (B \cup C) = (A \setminus B) \setminus C$.

$x \in A \wedge x \notin B \wedge x \notin C$ on both sides. $\checkmark$

(3) $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

$(x \in A \vee x \in B) \wedge x \notin C \iff (x \in A \wedge x \notin C) \vee (x \in B \wedge x \notin C)$. ✓

(4) $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.

Given $x \in A$, $\neg(x \in A \wedge x \in C)$ reduces to $x \notin C$; both sides are $x \in A \wedge x \in B \wedge x \notin C$. ✓

(5) $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$.

$x \in A \wedge (x \notin B \vee x \in C) \iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in C)$. ✓

(6) $(A \setminus B) \setminus C = (A \setminus C) \setminus (B \setminus C)$.

Given $x \notin C$, $B \setminus C$ reduces to B ; both sides are $x \in A \wedge x \notin B \wedge x \notin C$ (cf. Problem 22). ✓

(7) $A \triangle \emptyset = A$. $(A \setminus \emptyset) \cup (\emptyset \setminus A) = A \cup \emptyset = A$. ✓

(8) $A \triangle B = B \triangle A$. “In exactly one of A, B ” is symmetric in A, B . ✓

(9) $A \triangle (B \triangle C) = (A \triangle B) \triangle C$. Both sides hold iff x lies in an *odd* number of A, B, C (parity is associative). ✓

(10) $A \triangle A = \emptyset$. $(A \setminus A) \cup (A \setminus A) = \emptyset$. ✓

(11) $A \cap (B \triangle C) = (A \cap B) \triangle (A \cap C)$.

Both sides reduce to $(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in A \wedge x \notin B \wedge x \in C)$. ✓

Problem 55

Show $A \cup (B \setminus C) \neq (A \cup B) \setminus (A \cup C)$ in general.

Counterexample: $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$. LHS: $B \setminus C = \{2\}$, so $A \cup (B \setminus C) = \{1, 2\}$. RHS: $A \cup B = \{1, 2\}$, $A \cup C = \{1\}$, so $(A \cup B) \setminus (A \cup C) = \{2\}$. $\{1, 2\} \neq \{2\}$. □

Problem 56

Prove $A \triangle X = B$ has a unique solution.

The unique solution is $\boxed{X = A \triangle B}$.

Proof. Existence: Let $X = A \triangle B$. Then $A \triangle X = A \triangle (A \triangle B) = (A \triangle A) \triangle B = \emptyset \triangle B = B$. ✓

Uniqueness: If $A \triangle X = B$, apply $A \triangle$ to both sides: $A \triangle (A \triangle X) = A \triangle B$, giving $X = A \triangle B$. (We used associativity of $\triangle$ and $A \triangle A = \emptyset$, $\emptyset \triangle X = X$.) □

Problem 57

Do there exist sets A, B such that $|A \times B|$ equals:

57.1. 12 elements? **YES.** For example $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$; $|A| \cdot |B| = 3 \cdot 4 = 12$.

57.2. 13 elements? **YES.** Since 13 is prime, $|A| \cdot |B| = 13$ forces $|A| = 1, |B| = 13$ or vice versa. For example $A = \{1\}$, B any 13-element set.

$\boxed{57.1: \text{Yes (e.g., } 3 \times 4 = 12\text{)}. \quad 57.2: \text{Yes (e.g., } 1 \times 13 = 13\text{)}.$

Problem 58

Prove $A \times B = B \times A \implies A = B$ (assuming both non-empty).

Proof. Assume $A, B \neq \emptyset$ and $A \times B = B \times A$. To show $A \subseteq B$: let $a \in A$, pick any $b \in B$. Then $(a, b) \in A \times B = B \times A$, so $a \in B$. To show $B \subseteq A$: let $b \in B$, pick any $a \in A$. Then $(b, a) \in B \times A = A \times B$, so $b \in A$. Hence $A = B$. (If either set is empty, both products are

$\emptyset$ regardless of $A = B$.) □

Problem 59

List elements of $A \times B$ for $A = \{a, b\}$, $B = \{1, 2, 3, 4\}$.

$$A \times B = \{(a, 1), (a, 2), (a, 3), (a, 4), (b, 1), (b, 2), (b, 3), (b, 4)\}$$

(8 elements.)

Problem 60

Prove Cartesian product distributes over set operations.

(1) $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Proof. $(x, y) \in A \times (B \cup C) \iff x \in A \wedge (y \in B \vee y \in C) \iff (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C) \iff (x, y) \in (A \times B) \cup (A \times C)$. □

(2) $(B \cup C) \times A = (B \times A) \cup (C \times A)$. Same proof on the first component.

(3) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Proof. $(x, y) \in \text{LHS} \iff x \in A \wedge y \in B \wedge y \in C \iff (x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C)$. □

(4) $(B \cap C) \times A = (B \times A) \cap (C \times A)$. Mirror of (3).

(5) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

Proof. $(x, y) \in \text{LHS} \iff x \in A \wedge y \in B \wedge y \notin C$. $(x, y) \in \text{RHS} \iff (x \in A \wedge y \in B) \wedge \neg(x \in A \wedge y \in C)$. Given $x \in A$, the second condition reduces to $y \notin C$. □

(6) $(B \setminus C) \times A = (B \times A) \setminus (C \times A)$. Mirror of (5).

Problem 61

Prove $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$.

Proof. $(x, y) \in (A \cap B) \times (C \cap D) \iff x \in A \wedge x \in B \wedge y \in C \wedge y \in D \iff (x \in A \wedge y \in C) \wedge (x \in B \wedge y \in D) \iff (x, y) \in (A \times C) \cap (B \times D)$. □

Problem 62

Prove $A \subseteq B \wedge C \subseteq D \implies A \times C \subseteq B \times D$.

Proof. Let $(x, y) \in A \times C$. Then $x \in A \subseteq B$ gives $x \in B$, and $y \in C \subseteq D$ gives $y \in D$. So $(x, y) \in B \times D$. □

Practice 2–3: Relations

Problem 1.1

(a) Divisors of 12 in $\mathbb{N}$:

$$\{(1, 12), (2, 12), (3, 12), (4, 12), (6, 12), (12, 12)\}$$

(b) Pairs (d, n) with $d, n \in \{2, 3, 4, 5, 6\}$ and $d \mid n$:

$$\{(2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

(c) Triples (x, y, z) with $x = y + z$, all in $\{1, 2, 3\}$:

$$\{(2, 1, 1), (3, 1, 2), (3, 2, 1)\}$$

Problem 1.2

(a) $D = \{a, c, d\}, \quad R = \{1, 2, 4, 5\}.$

(b) Pattern $(n, 2n)$: $D = \mathbb{N}, \quad R = 2\mathbb{N}.$

(c) $x = y^2$: $D = [0, +\infty), \quad R = \mathbb{R}.$

(d) $x^2 + y^2 \leq 16$: $D = [-4, 4], \quad R = [-4, 4].$

(e) $(x, y) \in \mathbb{N}^2, \quad x \leq y$: $D = \mathbb{N}, \quad R = \mathbb{N}.$

(f) $(x, y) \in \mathbb{N}^2, \quad y \leq x$: $D = \mathbb{N}, \quad R = \mathbb{N}.$

(g) $x + y \geq 0$ on $\mathbb{R}^2$: $D = \mathbb{R}, \quad R = \mathbb{R}.$

(h) $2x \geq 3y$ on $\mathbb{R}^2$: $D = \mathbb{R}, \quad R = \mathbb{R}.$

Problem 2.1

On $A = \{a, b, c, d, e\}$.

(a) Symmetric; undirected edges: a – b , a – c , b – c , d – e .

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(b) Undirected edges: a – b , a – c , b – c , c – d . Vertex e isolated.

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(c) Identical to part (b); same graph and matrix.

Problem 2.2

On $A = \{a, b, c, d, e, f\}$. Rows/columns in order a, b, c, d, e, f .

(a) Directed edges: $a \rightarrow b, b \rightarrow c, a \rightarrow c, b \rightarrow e, c \rightarrow f, c \rightarrow d, d \rightarrow f, f \rightarrow e$.

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(b) Directed edges: $a \rightarrow b, b \rightarrow c, d \rightarrow c, d \rightarrow e, f \rightarrow e, f \rightarrow a, b \rightarrow e$.

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(c) Directed edges: $a \rightarrow c, c \rightarrow c, c \rightarrow d, c \rightarrow e, e \rightarrow c, e \rightarrow e, e \rightarrow f, a \rightarrow e$. Vertex b isolated.

$$M = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Problem 3.1

Proof sketch (standard properties of $^{-1}$ and $\circ$):

(i) $(\rho^{-1})^{-1} = \rho$: $(x, y) \in (\rho^{-1})^{-1} \Leftrightarrow (y, x) \in \rho^{-1} \Leftrightarrow (x, y) \in \rho$.

(ii) $(\rho \circ \sigma)^{-1} = \sigma^{-1} \circ \rho^{-1}$: $(z, x) \in (\rho \circ \sigma)^{-1} \Leftrightarrow \exists y: (x, y) \in \rho \wedge (y, z) \in \sigma \Leftrightarrow \exists y: (z, y) \in \sigma^{-1} \wedge (y, x) \in \rho^{-1} \Leftrightarrow (z, x) \in \sigma^{-1} \circ \rho^{-1}$. $\square$

Problem 3.2

$\rho = \{(x, y) \mid y = x^2 + 5\}$, $\sigma = \{(x, y) \mid y = 3x\}$ on $\mathbb{R}$. Composition is read left to right ($\rho \cdot \sigma$ applies ρ first).

$$\rho \cdot \sigma: x \xrightarrow{\rho} x^2 + 5 \xrightarrow{\sigma} 3(x^2 + 5). \quad \boxed{\rho \cdot \sigma = \{(x, 3x^2 + 15)\}}$$

$$\sigma \cdot \rho: x \xrightarrow{\sigma} 3x \xrightarrow{\rho} (3x)^2 + 5. \quad \boxed{\sigma \cdot \rho = \{(x, 9x^2 + 5)\}}$$

$$\boxed{\rho^{-1} = \{(x, y) \mid x = y^2 + 5\}} \text{ (i.e. } y = \pm\sqrt{x-5}, \text{ defined for } x \geq 5).$$

$$\boxed{\sigma^{-1} = \{(x, y) \mid y = x/3\}}.$$

$$\rho \cdot \sigma^{-1}: x \xrightarrow{\rho} x^2 + 5 \xrightarrow{\sigma^{-1}} \frac{x^2 + 5}{3}. \quad \boxed{\rho \cdot \sigma^{-1} = \left\{ \left(x, \frac{x^2 + 5}{3} \right) \right\}}$$

$$\rho^{-1} \cdot \sigma: x \xrightarrow{\rho^{-1}} \pm\sqrt{x-5} \xrightarrow{\sigma} \pm 3\sqrt{x-5} \quad (x \geq 5). \quad \boxed{\rho^{-1} \cdot \sigma = \{(x, \pm 3\sqrt{x-5}) \mid x \geq 5\}}.$$

Problem 3.3

$\rho = \{(x, x+2)\}$, $\sigma = \{(x, x^2)\}$ on $\mathbb{Z}$.

$\rho \cdot \sigma: x \xrightarrow{\rho} x+2 \xrightarrow{\sigma} (x+2)^2$. $\boxed{\rho \cdot \sigma = \{(x, (x+2)^2)\}}$.

$\sigma \cdot \rho: x \xrightarrow{\sigma} x^2 \xrightarrow{\rho} x^2+2$. $\boxed{\sigma \cdot \rho = \{(x, x^2+2)\}}$.

They are *not* equal (at $x = 1$: $\rho \cdot \sigma$ gives 9, $\sigma \cdot \rho$ gives 3).

Problem 3.4

$\rho = \{(x, y) \in \mathbb{Z}^2 \mid x+y \text{ is odd}\}$.

For ρ^2 : need z with $x+z$ odd and $z+y$ odd, so $x \equiv y \pmod{2}$. $\boxed{\rho^2 = \{(x, y) \in \mathbb{Z}^2 \mid x+y \text{ is even}\}}$.

$\boxed{\rho^3 = \rho}$, and in general $\rho^{2k} = \rho^2$, $\rho^{2k+1} = \rho$.

Problem 3.5

$\rho = \{(x, x+1) \mid x \in \mathbb{N}\}$.

$$\boxed{\rho^k = \{(x, x+k) \mid x \in \mathbb{N}\}, \quad k \geq 1.}$$

Problem 3.6

$\rho = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$.

$\boxed{\rho^2 = \{(1, 3), (2, 4), (3, 5)\}}$.

$\boxed{\rho^3 = \{(1, 4), (2, 5)\}}$.

$\boxed{\rho^4 = \{(1, 5)\}}$.

$\boxed{\rho^5 = \emptyset}$.

Problem 4.1

	Reflex.	Symm.	Antisymm.	Trans.
(a) $x^2 + y^2 = 1$, $\mathbb{R}$	No	Yes	No	No
(b) $x^2 = y^2$, $\mathbb{R}$	Yes	Yes	No	Yes
(c) x/y , $\mathbb{Z}$	Yes	No	No	Yes
(d) x/y , $\mathbb{N}$	Yes	No	Yes	Yes
(e) $\gcd = 1$, $\mathbb{N}$	No	Yes	No	No
(f) $\gcd = 1$, $\mathbb{Z}$	No	Yes	No	No

Here x/y denotes divisibility ($y \mid x$). (b) is an equivalence relation. (d) (divisibility on $\mathbb{N}$) is a partial order, but *not* total — e.g. 2 and 3 are incomparable. (c) (divisibility on $\mathbb{Z}$) is only a preorder: it fails antisymmetry, since $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$.

Problem 4.2

$\rho = \{(x, y) \in \mathbb{Z}^2 \mid |x - y| \text{ is even}\}$.

Reflexive – Yes; Symmetric – Yes; Antisymmetric – No; Transitive – Yes.

Equivalence relation. Classes: even integers and odd integers.

Problem 4.3

Preservation of **reflexivity** (R, S reflexive on A):

- (a) $R \cap S$: **True.** (b) $R \cup S$: **True.** (c) $R \circ S$: **True.**
 (d) $R \setminus S$: **False** ($(x, x) \in R \cap S$ so $(x, x) \notin R \setminus S$). (e) $R \oplus S$: **False** (same reason).

Problem 4.4

Preservation of **symmetry** (R, S symmetric):

- (a) $R \cap S$: **True.** (b) $R \cup S$: **True.**
 (c) $R \circ S$: **False.** ($R = \{(1, 2), (2, 1)\}$, $S = \{(2, 3), (3, 2)\}$: $(1, 3) \in R \circ S$ but $(3, 1) \notin R \circ S$.)
 (d) $R \setminus S$: **True.** (e) $R \oplus S$: **True.**

Problem 4.5

Preservation of **antisymmetry** (R, S antisymmetric):

- (a) $R \cap S$: **True.** (b) $R \cup S$: **False** ($R = \{(1, 2)\}$, $S = \{(2, 1)\}$).
 (c) $R \circ S$: **False.** (d) $R \setminus S$: **True** ($R \setminus S \subseteq R$).
 (e) $R \oplus S$: **False** (same counterexample as (b)).

Problem 4.6

Preservation of **transitivity** (R, S transitive):

- (a) $R \cap S$: **True.**
 (b) $R \cup S$: **False** ($R = \{(1, 2)\}$, $S = \{(2, 3)\}$; $(1, 3) \notin R \cup S$).
 (c) $R \circ S$: **False.**
 (d) $R \setminus S$: **False** ($R = \{(1, 2), (2, 3), (1, 3)\}$, $S = \{(1, 3)\}$; $R \setminus S = \{(1, 2), (2, 3)\}$, not transitive).
 (e) $R \oplus S$: **False.**

Problem 5.1

Construct a poset with a unique minimal element but no least element.

Note: In any *finite* poset a unique minimal element is automatically a least element, so an infinite poset (with an infinite descending chain) is required.

$A = \mathbb{Z}^- \cup \{m\}$, where $\mathbb{Z}^- = \{\dots, -3, -2, -1\}$ carries the usual order and m is incomparable to every element of $\mathbb{Z}^-$.

Then m is the *unique minimal* element: no $n \in \mathbb{Z}^-$ is minimal (the chain $\dots < -2 < -1$ descends forever, so each n has $n - 1 < n$ below it), and m has nothing below it. But m is *not* a least element, since $m \not\leq n$ for any $n \in \mathbb{Z}^-$ (m is incomparable to them). $\square$

Problem 5.2

Proof sketch. Let ρ be a partial order on A . Show ρ^{-1} is reflexive, antisymmetric, transitive. Reflexive: $(x, x) \in \rho \Rightarrow (x, x) \in \rho^{-1}$. Antisymmetric: $(x, y), (y, x) \in \rho^{-1} \Rightarrow (y, x), (x, y) \in \rho \Rightarrow x = y$. Transitive: $(x, y), (y, z) \in \rho^{-1} \Rightarrow (z, y), (y, x) \in \rho \Rightarrow (z, x) \in \rho \Rightarrow (x, z) \in \rho^{-1}$. $\square$

Problem 5.3

Proof sketch. Let $\{\rho_i\}$ be partial orders on A , $\rho = \bigcap_i \rho_i$. Reflexive: $(x, x) \in \rho_i$ for all i . Antisymmetric: $(x, y), (y, x) \in \rho \subseteq \rho_i \Rightarrow x = y$. Transitive: $(x, y), (y, z) \in \rho \subseteq \rho_i \Rightarrow (x, z) \in \rho_i$ for all $i \Rightarrow (x, z) \in \rho$. $\square$

Problem 5.4

Proof sketch. $\rho_A = \rho \cap (A \times A)$ inherits reflexivity (for $x \in A$), antisymmetry, and transitivity from ρ since $\rho_A \subseteq \rho$ and all elements in consideration belong to A . $\square$

Problem 5.5

$\rho_1 \cup \rho_2$ is a linear order iff $\rho_1 \subseteq \rho_2$ or $\rho_2 \subseteq \rho_1$.

If one contains the other, the union equals the larger one. Conversely, if neither contains the other, antisymmetry of the union can fail.

Problem 5.6

Proof sketch. Induction on $|A| = n$. Base $n = 1$: trivial. Step: pick $a \in A$; by hypothesis $A \setminus \{a\}$ has a linear order $a_1 < \dots < a_{n-1}$. Insert a at the end: $a_1 < \dots < a_{n-1} < a$. This is a linear order on A . $\square$

Problem 6.1

(a) $xy > 0$ on $\mathbb{Q}$: **No.** Not reflexive ($0 \cdot 0 = 0 \not> 0$).

(b) $x - y \in \mathbb{Z}$ on $\mathbb{Q}$: **Yes.** Reflexive ($x - x = 0 \in \mathbb{Z}$), symmetric, and transitive ($x - y, y - z \in \mathbb{Z} \Rightarrow x - z \in \mathbb{Z}$). The classes are the cosets $x + \mathbb{Z}$ — rationals sharing the same fractional part.

(c) $(a \leq b) \vee (a > b)$ on $\mathbb{N}$: **Yes.** By trichotomy the predicate always holds, so $\rho = \mathbb{N} \times \mathbb{N}$ is the universal relation — trivially reflexive, symmetric, transitive (a single class, all of $\mathbb{N}$).

(d) $|a - b| \leq 2$ on $\mathbb{N}$: **No.** Not transitive ($|1 - 3| = 2 \leq 2$, $|3 - 5| = 2 \leq 2$, but $|1 - 5| = 4 > 2$).

Problem 6.2

(Equivalence relation $\Leftrightarrow$ partition.)

Proof sketch. ($\Rightarrow$) Classes cover A (reflexivity: $x \in [x]$). If $[a] \cap [b] \neq \emptyset$, pick c in both; then apc and cpb , so apb and $[a] = [b]$. ($\Leftarrow$) Define xpy iff x, y are in the same block; reflexivity, symmetry, transitivity are immediate. $\square$

Problem 6.3 $\mathbb{Z}_2$:

+	[0]	[1]
[0]	[0]	[1]
[1]	[1]	[0]

·	[0]	[1]
[0]	[0]	[0]
[1]	[1]	[1]

 $\mathbb{Z}_3$:

+	[0]	[1]	[2]
[0]	[0]	[1]	[2]
[1]	[1]	[2]	[0]
[2]	[2]	[0]	[1]

·	[0]	[1]	[2]
[0]	[0]	[0]	[0]
[1]	[1]	[1]	[2]
[2]	[2]	[2]	[1]

 $\mathbb{Z}_4$:

+	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]
[1]	[1]	[2]	[3]	[0]
[2]	[2]	[3]	[0]	[1]
[3]	[3]	[0]	[1]	[2]

·	[0]	[1]	[2]	[3]
[0]	[0]	[0]	[0]	[0]
[1]	[1]	[0]	[1]	[3]
[2]	[2]	[0]	[2]	[2]
[3]	[3]	[0]	[3]	[1]

Problem 6.4 $[a] \in \mathbb{Z}_n$ is invertible iff $\gcd(a, n) = 1$.

Proof sketch. $(\Rightarrow)$ $ab \equiv 1 \pmod{n}$ means $ab - kn = 1$, so $\gcd(a, n) = 1$ by Bézout. $(\Leftarrow)$ $\gcd(a, n) = 1$ implies $\exists b, k: ab + kn = 1$, so $ab \equiv 1 \pmod{n}$. $\square$

Problem 7.1On $A = \{a, b, c\}$, $\Delta_A = \{(a, a), (b, b), (c, c)\}$.

(a) $\rho = \{(a, b), (b, a)\}$. $r(\rho) = \rho \cup \Delta_A = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$. $s(\rho) = \rho$ (already symmetric). $t(\rho) = \{(a, b), (b, a), (a, a), (b, b)\}$.

(b) $\rho = \{(a, b), (b, c)\}$. $r(\rho) = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$. $s(\rho) = \{(a, b), (b, a), (b, c), (c, b)\}$. $t(\rho) = \{(a, b), (b, c), (a, c)\}$.

(c) $\rho = \{(a, a), (a, b), (c, b), (c, a)\}$. $r(\rho) = \rho \cup \{(b, b), (c, c)\}$. $s(\rho) = \rho \cup \{(b, a), (b, c), (a, c)\}$. $t(\rho) = \rho$ (already transitive: $(c, a) \circ (a, b) = (c, b) \in \rho$).

(d) $\rho = \{(a, a), (a, c), (b, c)\}$. $r(\rho) = \rho \cup \{(b, b), (c, c)\}$. $s(\rho) = \rho \cup \{(c, a), (c, b)\}$. $t(\rho) = \rho$ (already transitive).

(e) $\rho = \{(a, b), (c, a), (b, c)\}$. $r(\rho) = \rho \cup \Delta_A$. $s(\rho) = \{(a, b), (b, a), (c, a), (a, c), (b, c), (c, b)\}$. $t(\rho) = A \times A$ (the full cycle $a \rightarrow b \rightarrow c \rightarrow a$ generates all 9 pairs).

(f) $\rho = \{(a, b), (c, b), (b, c)\}$. $r(\rho) = \rho \cup \Delta_A$. $s(\rho) = \{(a, b), (b, a), (c, b), (b, c)\}$. $t(\rho) = \{(a, b), (a, c), (b, c), (b, b), (c, b), (c, c)\}$.

Problem 7.2 $A = \{1, 2, 3, 4\}$, $\rho = \{(1, 2), (3, 1), (3, 2), (2, 4)\}$.

$$r(\rho) = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 4), (3, 1), (3, 2)\}.$$

$$s(\rho) = \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2), (2, 4), (4, 2)\}.$$

$$t(\rho) = \{(1, 2), (1, 4), (2, 4), (3, 1), (3, 2), (3, 4)\}.$$

(New pairs from ρ^2 : $(1, 4)$ via $1 \rightarrow 2 \rightarrow 4$; $(3, 4)$ via $3 \rightarrow 2 \rightarrow 4$; $(3, 2)$ via $3 \rightarrow 1 \rightarrow 2$ already present. Vertex 4 is a sink, so ρ^3 adds nothing.)

Problem 7.3

Proof sketch. $\tau = \rho \cap \sigma$ where ρ, σ are equivalence relations. Reflexive: $(x, x) \in \rho \cap \sigma$. Symmetric: $(x, y) \in \tau \Rightarrow (y, x) \in \rho$ and $(y, x) \in \sigma \Rightarrow (y, x) \in \tau$. Transitive: $(x, y), (y, z) \in \tau \subseteq \rho \Rightarrow (x, z) \in \rho$; similarly $\in \sigma$; so $(x, z) \in \tau$. $\square$

Problem 7.4

$\rho \circ \theta$ is an equivalence relation iff $\rho \circ \theta = \theta \circ \rho$.

Proof sketch. $(\Rightarrow)$ $(\rho \circ \theta)^{-1} = \theta^{-1} \circ \rho^{-1} = \theta \circ \rho$. Since $\rho \circ \theta$ is symmetric, $\rho \circ \theta = \theta \circ \rho$.

$(\Leftarrow)$ Reflexivity: $\Delta \subseteq \rho, \theta \Rightarrow \Delta \subseteq \rho \circ \theta$. Symmetry: $(\rho \circ \theta)^{-1} = \theta \circ \rho = \rho \circ \theta$. Transitivity: $(\rho \circ \theta)^2 = \rho \circ (\theta \circ \rho) \circ \theta = \rho \circ (\rho \circ \theta) \circ \theta \subseteq \rho^2 \circ \theta^2 \subseteq \rho \circ \theta$. $\square$

Problem 7.5

$\rho \cup \theta$ is an equivalence relation iff $\rho \circ \theta = \theta \circ \rho$.

Proof sketch. $(\Rightarrow)$ If $\rho \cup \theta$ is an equivalence (hence transitive), then $\rho \circ \theta \subseteq (\rho \cup \theta) \circ (\rho \cup \theta) \subseteq \rho \cup \theta$. Similarly $\theta \circ \rho \subseteq \rho \cup \theta$. Since $\rho \cup \theta \subseteq \rho \circ \theta$ (as $\rho = \rho \circ \Delta \subseteq \rho \circ \theta$), we get $\rho \cup \theta = \rho \circ \theta$, and by symmetry also $= \theta \circ \rho$.

$(\Leftarrow)$ If $\rho \circ \theta = \theta \circ \rho$, then by Problem 7.4, $\rho \circ \theta$ is an equivalence relation. One shows $\rho \cup \theta = \rho \circ \theta$, which is then an equivalence relation. $\square$

Practice 4: Functions (Mappings)

Problem 1

f is total and g is partial. Composition is left to right, $(f \cdot g)(x) = g(f(x))$:

- $g \cdot f$ (apply g first): defined at x only if $g(x)$ is defined; wherever $g(x)$ is defined, $f(g(x))$ is defined since f is total. So $\text{Dom}(g \cdot f) = \text{Dom}(g)$.
- $f \cdot g$ (apply f first): $f(x)$ is always defined, but $g(f(x))$ requires $f(x) \in \text{Dom}(g)$. So $f \cdot g$ may be partial: defined at x only if $f(x) \in \text{Dom}(g)$.

Problem 2

- $f(x) = x^2 + 3$: $D_f = \mathbb{R}$, $R_f = [3, +\infty)$.
- $f(x) = \sqrt{x-2}$: $D_f = [2, +\infty)$, $R_f = [0, +\infty)$.
- $f(x) = \frac{1}{\sqrt{x-2}}$: $D_f = (2, +\infty)$, $R_f = (0, +\infty)$.
- $f(x) = |x|$: $D_f = \mathbb{R}$, $R_f = [0, +\infty)$.
- $f(x) = \frac{1}{x^2+2}$: $D_f = \mathbb{R}$, $R_f = (0, \frac{1}{2}]$.
- $f(x) = \frac{1}{x^2-2}$: $D_f = \mathbb{R} \setminus \{-\sqrt{2}, \sqrt{2}\}$, $R_f = (-\infty, -\frac{1}{2}] \cup (0, +\infty)$.

Problem 3

- $\rho = \{(x, \sqrt{x^2+4}) \mid x \in \mathbb{R}\}$: **Yes**, it is a function. Each x maps to a unique value $\sqrt{x^2+4}$.
- $\rho = \{(x, 5) \mid x \in \mathbb{R}\}$: **Yes**, it is a (constant) function.
- $\rho = \{(7, y) \mid y \in \mathbb{R}\}$: **No**, it is not a function. The element 7 is paired with every real number, violating single-valuedness.

Problem 4

(Left-to-right convention: $(f \cdot g)(x) = g(f(x))$, apply f first.)

- $f(x) = x^2 + 1$, $g(x) = x + 3$:

$$(f \cdot g)(x) = x^2 + 4, \quad (g \cdot f)(x) = x^2 + 6x + 10.$$
- $f(x) = \sqrt{x^2+2}$, $g(x) = x^2 + 3$:

$$(f \cdot g)(x) = x^2 + 5, \quad (g \cdot f)(x) = \sqrt{x^4 + 6x^2 + 11}.$$
- $f(x) = \frac{1}{x}$, $g(x) = 2x + 3$:

$$(f \cdot g)(x) = \frac{2+3x}{x} \ (x \neq 0), \quad (g \cdot f)(x) = \frac{1}{2x+3} \ (x \neq -\frac{3}{2}).$$

Problem 5

- $f(x) = \frac{x+4}{2}$: $f^{-1}(x) = 2x - 4$.
- $f(x) = x^3$: $f^{-1}(x) = \sqrt[3]{x}$.
- $f(x) = \frac{x-2}{x+3}$: $f^{-1}(x) = \frac{2+3x}{1-x}$, $x \neq 1$.

Problem 6

- $f(x) = |x|$: neither injective nor surjective. ($f(1) = f(-1)$; range = $[0, \infty) \neq \mathbb{R}$.)

- (b) $f(x) = x^2 + 4$: neither injective nor surjective. ($f(1) = f(-1)$; range $= [4, \infty) \neq \mathbb{R}$.)
 (c) $f(x) = x^3 + 6$: **bijective**. (Strictly increasing, hence injective; cube root gives surjectivity.)
 (d) $f(x) = x + |x|$: neither injective nor surjective. ($f(x) = 0$ for all $x < 0$; range $= [0, \infty) \neq \mathbb{R}$.)
 (e) $f(x) = x(x-2)(x+2) = x^3 - 4x$: surjective but **not** injective. (Cubic $\rightarrow \pm\infty$ gives surjectivity; $f(0) = f(2) = 0$ shows non-injectivity.)

Problem 7

The kernel $\text{Ker}(f) = \{(x_1, x_2) : f(x_1) = f(x_2)\}$. The equivalence classes are:

- (a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = |x|$: $\text{Ker}(f)$ partitions $\mathbb{Z}$ into $\{0\}$ and $\{-n, n\}$ for each $n \geq 1$.
 (b) $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lfloor x \rfloor$: classes are the half-open intervals $[n, n+1)$ for each $n \in \mathbb{Z}$.
 (c) $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lceil x \rceil$: classes are the half-open intervals $(n-1, n]$ for each $n \in \mathbb{Z}$.

Problem 8

We must show $(a) \Leftrightarrow (b)$, where (a) is $f(x) = f(y) \Rightarrow x = y$ and (b) is $x \neq y \Rightarrow f(x) \neq f(y)$. Statement (b) is the contrapositive of (a). A conditional and its contrapositive are logically equivalent, so $(a) \Leftrightarrow (b)$. $\square$

Problem 9

(Reference: Lemmas 1, 2, 3, 5 from the lecture notes on functions.)

Lemma 1 (Composition of injections is injective): If f, g injective, let $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $g(f(x_1)) = g(f(x_2))$; g injective $\Rightarrow f(x_1) = f(x_2)$; f injective $\Rightarrow x_1 = x_2$. $\square$

Lemma 2 (Composition of surjections is surjective): If $f : A \rightarrow B$ and $g : B \rightarrow C$ are surjective, given $c \in C$, $\exists b \in B$ with $g(b) = c$, and $\exists a \in A$ with $f(a) = b$, so $(g \circ f)(a) = c$. $\square$

Lemma 3 (Composition of bijections is bijective): Follows from Lemmas 1 and 2.

Lemma 5 (If $g \circ f$ is injective then f is injective): Suppose $f(x_1) = f(x_2)$. Then $g(f(x_1)) = g(f(x_2))$, i.e. $(g \circ f)(x_1) = (g \circ f)(x_2)$; since $g \circ f$ is injective, $x_1 = x_2$. $\square$

Problem 10

$\text{Ker}(f) = \{(a_1, a_2) \in A \times A : f(a_1) = f(a_2)\}$ is an equivalence relation:

- *Reflexive*: $f(a) = f(a)$, so $a \sim a$.
- *Symmetric*: $f(a) = f(b) \Rightarrow f(b) = f(a)$.
- *Transitive*: $f(a) = f(b)$ and $f(b) = f(c) \Rightarrow f(a) = f(c)$. $\square$

Problem 11

- (a) $f : \mathbb{Z} \rightarrow \mathbb{N}$, $f(x) = |x|$: equivalence classes are $\{0\}$ and $\{-n, n\}$ for $n \geq 1$.
 (b) $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lfloor x \rfloor$: equivalence classes are the intervals $[n, n+1)$ for each $n \in \mathbb{Z}$.

(Same as Problem 7(a) and 7(b).)

Problem 12

$f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$, $f(x) = \frac{x}{2x+1}$.

Injective: $\frac{x_1}{2x_1+1} = \frac{x_2}{2x_2+1} \Rightarrow$ cross-multiply $\Rightarrow 2x_1x_2 + x_1 = 2x_1x_2 + x_2 \Rightarrow x_1 = x_2$. ✓

Not surjective: $f(x) = \frac{1}{2+1/x} < \frac{1}{2}$ for all $x > 0$, so $y = \frac{1}{2} \in \mathbb{Q}^+$ has no preimage. ✓

Problem 13

$f : \mathbb{Z}_{15} \rightarrow \mathbb{Z}_{15}, f(x) \equiv 4x \pmod{15}$.

Bijjective: $\gcd(4, 15) = 1$, so multiplication by 4 is a bijection on $\mathbb{Z}_{15}$.

Inverse: We need $4^{-1} \pmod{15}$. Since $4 \cdot 4 = 16 \equiv 1 \pmod{15}$, we have $4^{-1} \equiv 4$. Thus $f^{-1}(x) \equiv 4x \pmod{15}$ (the function is its own inverse).

Problem 14

$f : \mathbb{Z}_{15} \rightarrow \mathbb{Z}_{15}, f(x) \equiv 2x + 4 \pmod{15}$.

Bijjective. Since $\gcd(2, 15) = 1$, multiplication by 2 is invertible mod 15, so f is a bijection. Explicit check: $f(0) = 4, f(1) = 6, f(2) = 8, \dots, f(13) = 0, f(14) = 2$ — all 15 residues appear.

Inverse: $2^{-1} \equiv 8 \pmod{15}$ (since $2 \cdot 8 = 16 \equiv 1$). So $f^{-1}(y) \equiv 8(y - 4) \equiv 8y + 13 \pmod{15}$.

Problem 15

Claim: If A is finite, then $f : A \rightarrow A$ is bijective iff it is injective.

Proof sketch: $(\Rightarrow)$ Bijective $\Rightarrow$ injective trivially. $(\Leftarrow)$ If f is injective, then $|f(A)| = |A|$. Since $f(A) \subseteq A$ and A is finite, $|f(A)| = |A|$ forces $f(A) = A$, so f is surjective, hence bijective. $\square$

Problem 16

Claim: If A is finite, then $f : A \rightarrow A$ is bijective iff it is surjective.

Proof sketch: $(\Rightarrow)$ Trivial. $(\Leftarrow)$ If f is surjective, then $f(A) = A$, so $|f(A)| = |A|$. By the pigeonhole principle, a surjection from a finite set to itself must be injective (otherwise two inputs share an output, leaving at most $|A| - 1$ distinct outputs, contradicting surjectivity). $\square$

Problem 17

Corollary 1 (from lecture): For a finite set A , $f : A \rightarrow A$ is injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective.

This is the direct combination of Problems 15 and 16. $\square$

Problem 18

(Reference: Theorems 6 and 7 from lecture notes.)

Theorem 6 (f has a left inverse iff f is injective): $(\Leftarrow)$ If $f : A \rightarrow B$ is injective, define $g : B \rightarrow A$ by $g(y) =$ the unique x with $f(x) = y$ if $y \in \text{Ran}(f)$, and $g(y) = a_0$ (fixed element) otherwise. Then $g \circ f = \text{id}_A$. $(\Rightarrow)$ If $g \circ f = \text{id}_A$, then $f(x_1) = f(x_2) \Rightarrow g(f(x_1)) = g(f(x_2)) \Rightarrow x_1 = x_2$.

Theorem 7 (f has a right inverse iff f is surjective): $(\Leftarrow)$ If f is surjective, for each $y \in B$ choose some $x_y \in f^{-1}(\{y\})$ and define $h(y) = x_y$. Then $f \circ h = \text{id}_B$. $(\Rightarrow)$ If $f \circ h = \text{id}_B$, then

for any $y \in B$, $f(h(y)) = y$, so $y \in \text{Ran}(f)$.

Problem 19

$f(A \cap B) = f(A) \cap f(B)$ holds for all A, B if and only if f is **injective**.

Sketch: ($\Rightarrow$) If f not injective, take $A = \{x_1\}$, $B = \{x_2\}$ with $x_1 \neq x_2$, $f(x_1) = f(x_2)$: then $f(\emptyset) = \emptyset \neq \{f(x_1)\} = f(A) \cap f(B)$. ($\Leftarrow$) If f injective, the reverse inclusion $f(A) \cap f(B) \subseteq f(A \cap B)$ holds because $f(a) = f(b) \Rightarrow a = b$. $\square$

Problem 20

$f(A \setminus B) = f(A) \setminus f(B)$ holds for all A, B if and only if f is **injective**.

Sketch: ($\Rightarrow$) If f not injective, take $a \neq b$ with $f(a) = f(b)$, $A = \{a, b\}$, $B = \{b\}$: then $f(A \setminus B) = f(\{a\}) = \{f(a)\}$ but $f(A) \setminus f(B) = \{f(a)\} \setminus \{f(b)\} = \emptyset$. ($\Leftarrow$) If f injective and $y \in f(A \setminus B)$, then $y = f(a)$ for unique $a \in A \setminus B$; injectivity ensures no $b \in B$ maps to y , so $y \notin f(B)$, giving $y \in f(A) \setminus f(B)$. The other inclusion holds for any function. $\square$

Problem 21

Yes. It is possible for $X_1 \neq X_2$ but $f(X_1) = f(X_2)$.

Example: $f : \{1, 2\} \rightarrow \{a\}$ with $f(1) = f(2) = a$. Take $X_1 = \{1\}$, $X_2 = \{1, 2\}$. Then $X_1 \neq X_2$ but $f(X_1) = \{a\} = f(X_2)$.

Problem 22

Yes. Disjoint non-empty subsets can have equal images when f is not injective.

Example: $A = \{1, 2, 3, 4\}$, $B = \{a, b\}$, $f(1) = a$, $f(2) = b$, $f(3) = a$, $f(4) = b$. Take $X_1 = \{1, 2\}$, $X_2 = \{3, 4\}$. Then $X_1 \cap X_2 = \emptyset$ and $f(X_1) = \{a, b\} = f(X_2)$.

Problem 23

Yes. It is possible for $Y_1 \neq Y_2$ but $f^{-1}(Y_1) = f^{-1}(Y_2)$.

Example: $f : \{1\} \rightarrow \{a, b\}$ with $f(1) = a$. Let $Y_1 = \{a\}$ and $Y_2 = \{a, b\}$. Then $Y_1 \neq Y_2$ but $f^{-1}(Y_1) = \{1\} = f^{-1}(Y_2)$, because b has no preimage.

Problem 24

Yes, it is always true. If $Y_1 \cap Y_2 = \emptyset$, then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \emptyset$.

Proof: $f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2) = f^{-1}(\emptyset) = \emptyset$, using the fact that preimages preserve intersections. $\square$

Problem 25

Claim: $f(f^{-1}(Y)) \subseteq Y$ for any $f : A \rightarrow B$ and $Y \subseteq B$.

Proof: Let $y \in f(f^{-1}(Y))$. Then $y = f(x)$ for some $x \in f^{-1}(Y)$, meaning $f(x) \in Y$, so $y \in Y$. $\square$

Counterexample to equality: $A = \{1\}$, $B = \{a, b\}$, $f(1) = a$, $Y = B$. Then $f(f^{-1}(Y)) = \{a\} \subsetneq \{a, b\} = Y$.

Equality holds for all $Y \subseteq B$ iff f is surjective.

Practice 5: Combinatorics — Basic Rules

Section 1: Sum and Product Rules

Problem 1

Count of integers from 0 to 1000 whose digits contain *exactly one* 6.

Write each integer 0–999 as a 3-digit string with leading zeros (leading zeros add no 6's). Exactly one of the three digits is a 6: choose its position (3 ways), set it to 6, and fill the other two with non-6 digits (9 each). The number 1000 has no 6.

$$\binom{3}{1} \cdot 9^2 = 3 \cdot 81 = \boxed{243}$$

Problem 2

Count of integers from 0 to 1000 whose digits contain *one or more* 6 (at least one).

Complement: count integers with *no* digit 6.

- 1-digit numbers (0–9): 9 have no 6 (all except 6).
- 2-digit numbers (10–99): first digit from $\{1, \dots, 9\} \setminus \{6\} = 8$, second digit from $\{0, \dots, 9\} \setminus \{6\} = 9$. Total $8 \cdot 9 = 72$.
- 3-digit numbers (100–999): $8 \cdot 9 \cdot 9 = 648$.
- The number 1000 contains no 6: 1 more.

No-six total: $9 + 72 + 648 + 1 = 730$; all integers from 0 to 1000: 1001.

$$\boxed{1001 - 730 = 271}$$

Problem 3

Choose two books of different subjects from 15 Informatics, 12 Math, 10 Physics.

By the sum/product rules: $15 \cdot 12 + 15 \cdot 10 + 12 \cdot 10 = 180 + 150 + 120 = \boxed{450}$.

Problem 4

Odd numbers between 100 and 1000 (i.e. 3-digit odd numbers 101, 103, ..., 999).

Last digit: 5 choices ($\{1, 3, 5, 7, 9\}$). First digit: 9 choices (1–9). Middle digit: 10 choices (0–9).

$$\boxed{9 \times 10 \times 5 = 450}$$

Problem 5

Natural numbers less than 700 ($\{1, \dots, 699\}$):

(a) Divisible by 5: $\lfloor 699/5 \rfloor = \boxed{139}$.

(b) Of those (divisible by 5), how many are also divisible by 3? They are exactly the multiples of $\text{lcm}(3, 5) = 15$: $\lfloor 699/15 \rfloor = \boxed{46}$.

(c) Of those, how many are divisible by both 3 and 5 (i.e. by 15): $\lfloor 699/15 \rfloor = \boxed{46}$.

Problem 6

Yerevan $\rightarrow$ Moscow: 15 flights. Moscow $\rightarrow$ London: 20 flights. By the product rule:
 $15 \times 20 = 300$.

Problem 7

4-digit numbers with at least two identical digits.

Total 4-digit numbers: $9 \times 10^3 = 9000$. 4-digit numbers with all digits distinct: first digit 9 choices, second 9, third 8, fourth 7, giving $9 \times 9 \times 8 \times 7 = 4536$.

$$9000 - 4536 = 4464$$

Problem 8

7-digit numbers where first and last digits are different.

Total 7-digit numbers: 9×10^6 . 7-digit numbers where first = last digit: first digit 9 choices, last digit fixed equal to first (1 choice), middle 5 digits 10^5 each. Total same: 9×10^5 .

Answer: $9 \times 10^6 - 9 \times 10^5 = 9 \times 10^5(10 - 1) = 9 \times 10^5 \times 9$.

$$9 \times 10^6 - 9 \times 10^5 = 8,100,000$$

Problem 9

6-digit numbers where adjacent digits are different (digits 0–9, leading digit $\neq 0$).

Position 1: 9 choices (1–9). Each subsequent position: 9 choices (any digit except the preceding one).

$$9 \times 9^5 = 9^6 = 531,441$$

Problem 10

7-digit numbers: even positions (2,4,6) hold odd digits; odd positions (1,3,5,7) hold even digits.

Positions 2,4,6: 5 odd digits each $\Rightarrow 5^3$. Position 1 (leading, must be even and $\neq 0$): 4 choices {2, 4, 6, 8}. Positions 3,5,7: 5 even digits each $\Rightarrow 5^3$.

$$4 \times 5^3 \times 5^3 = 4 \times 15,625 = 62,500$$

Section 2: Permutations and Arrangements**Problem 1**

Mappings from a 6-element set to a 3-element set: each of the 6 elements maps to one of 3 images.

$$3^6 = 729$$

Problem 2

Injective mappings from a 6-element set to a 3-element set. Since $6 > 3$, by the Pigeonhole Principle no injection exists.

$$0$$

Problem 3

5 boys and 6 girls in a line; boys occupy the first 5 positions.

Arrange 5 boys in the first 5 positions: $5!$. Arrange 6 girls in the remaining 6 positions: $6!$.

$$5! \times 6! = 120 \times 720 = 86,400$$

Problem 4

6 boys and 7 girls (13 people) in a line; first position is a boy.

Choose the boy for position 1: 6 ways. Arrange remaining 12 people: $12!$.

$$6 \times 12! = 6 \times 479,001,600 = 2,874,009,600$$

Problem 5

6 boys and 7 girls in a line; first and last positions are boys.

First position: 6 choices. Last position: 5 remaining boys. Middle 11 positions: $11!$.

$$6 \times 5 \times 11! = 30 \times 39,916,800 = 1,197,504,000$$

Problem 6

5 boys and 6 girls in a line; boys occupy 5 *consecutive* positions.

Treat the block of 5 boys as a single unit. Together with 6 girls, we have 7 objects to arrange in a line: $7!$ ways. Within the block the 5 boys can be permuted: $5!$ ways.

$$7! \times 5! = 5040 \times 120 = 604,800$$

Problem 7

Arrange 9 distinct people in a line.

$$9! = 362,880$$

Problem 8

8-digit numbers formed from the digits of 23122313.

Digits: 1 appears 2 times, 2 appears 3 times, 3 appears 3 times. Total 8 digits.

Permutations with repetition: $\frac{8!}{2! 3! 3!} = \frac{40320}{2 \cdot 6 \cdot 6} = \frac{40320}{72}$.

$$\frac{8!}{2! 3! 3!} = 560$$

Problem 9

Words of length 6 from alphabet $\{a, b, g, d, e, z\}$ (vowels: a, e ; consonants: b, g, d, z) where two vowels are adjacent.

Total words of length 6: $6^6 = 46656$.

Strings with NO two adjacent vowels: Let V_n, C_n = strings ending in vowel/consonant. $V_n = 2C_{n-1}$, $C_n = 4(V_{n-1} + C_{n-1})$.

$V_1 = 2, C_1 = 4, V_2 = 8, C_2 = 24, V_3 = 48, C_3 = 128, V_4 = 256, C_4 = 704, V_5 = 1408, C_5 =$

3840, $V_6 = 7680$, $C_6 = 20992$. Total: $f(6) = 28672$.

$$46656 - 28672 = 17984$$

Problem 10

Prove: In any group of 6 people, there are either 3 mutual friends or 3 mutual strangers ($R(3, 3) = 6$).

Proof sketch. Model the 6 people as vertices of K_6 with edges coloured red (friend) or blue (stranger). Pick any vertex A ; it has 5 edges, so by the Pigeonhole Principle at least 3 share one colour, say red, to vertices B, C, D . If any edge among B, C, D is red, it forms a red triangle with A ; otherwise B, C, D form a blue triangle. $\square$

Problem 11

How many numbers must be selected from $\{1, 2, \dots, 210\}$ to guarantee at least one even number?

There are 105 odd numbers in $\{1, \dots, 210\}$. In the worst case we pick all 105 odd numbers first. One more pick guarantees an even number.

$$106$$

Problem 12

Prove: Among the first 2014 terms of $7, 77, 777, \dots$, one is divisible by 2013.

Proof sketch. Let $a_k = \underbrace{77 \dots 7}_k$. Consider the 2014 remainders $a_k \bmod 2013$. If some remainder is 0, done. Otherwise 2014 remainders fall into $\{1, \dots, 2012\}$ (2012 classes), so by Pigeonhole two terms a_i, a_j ($i < j$) share a remainder. Then $2013 \mid a_j - a_i = a_{j-i} \cdot 10^i$. Since $\gcd(10, 2013) = 1$, we get $2013 \mid a_{j-i}$, and $1 \leq j - i \leq 2013$. $\square$

Practice 6: Combinations and Binomial Coefficients

Problem 1

Bit strings of length 9 with exactly 5 zeros (and 4 ones).

Choose positions for the 5 zeros: $\binom{9}{5}$.

$$\binom{9}{5} = 126$$

Problem 2

Committee of 6 teachers and 8 students from 12 teachers and 20 students.

$$\binom{12}{6} \cdot \binom{20}{8} = 924 \times 125970 = 116,396,280$$

Problem 3

Choose 13 cards from 52 such that they include 5 cards of the same suit.

Using complement (no suit has ≥ 5 cards, meaning each suit contributes ≤ 4 cards):

The possible distributions (a, b, c, d) with each ≤ 4 summing to 13 are: $(4, 4, 4, 1)$, $(4, 4, 3, 2)$, $(4, 3, 3, 3)$.

$$(4, 4, 4, 1) : 4 \cdot \binom{13}{4}^3 \binom{13}{1} = 19,007,345,500$$

$$(4, 4, 3, 2) : 12 \cdot \binom{13}{4}^2 \binom{13}{3} \binom{13}{2} = 136,852,887,600$$

$$(4, 3, 3, 3) : 4 \cdot \binom{13}{4} \binom{13}{3}^3 = 66,905,856,160$$

$N_{\leq 4} = 222,766,089,260$. Total 13-card hands: $\binom{52}{13} = 635,013,559,600$.

$$635,013,559,600 - 222,766,089,260 = 412,247,470,340$$

Problem 4

Divide 10 people into two teams of 5.

Since the two teams are unordered (indistinguishable labels): $\frac{1}{2} \binom{10}{5} = \frac{252}{2}$.

$$126$$

Problem 5

$A = \{a, b, c, d, e, f, g, h\}$. Number of 5-element subsets containing element a .

Fix a in the subset; choose 4 more from the remaining 7 elements.

$$\binom{7}{4} = 35$$

Problem 6

Coefficient of x^6y^4 in $(2x + 3y)^{10}$.

By the binomial theorem: $\binom{10}{6}(2x)^6(3y)^4 = \binom{10}{6} \cdot 2^6 \cdot 3^4 \cdot x^6y^4$.

$$\binom{10}{6} = 210, 2^6 = 64, 3^4 = 81.$$

$$210 \cdot 64 \cdot 81 = 1,088,640$$

Problem 7

Rook at a1 reaching e5, moving only right or up.

From a1 to e5: 4 steps right ($a \rightarrow b \rightarrow c \rightarrow d \rightarrow e$) and 4 steps up ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$).

Total 8 steps, choose 4 for right.

$$\binom{8}{4} = 70$$

Problem 8

Lattice paths from $O(0, 0)$ to $A(7, 8)$ through $(3, 4)$.

$O \rightarrow P(3, 4)$: $\binom{7}{3} = 35$ paths. $P \rightarrow A$: $\binom{8}{4} = 70$ paths.

$$35 \times 70 = 2450$$

Problem 9

Number of ways to write 6 as an ordered sum of three non-negative summands, $x_1 + x_2 + x_3 = 6$ (this is in the sheet's “combinations with repetition” section). By stars and bars:

$$\binom{6 + 3 - 1}{3 - 1} = \binom{8}{2} = \boxed{28}.$$

(If the parts are required to be ≥ 1 it is $\binom{5}{2} = 10$; the *unordered* partitions $4+1+1$, $3+2+1$, $2+2+2$ number 3.)

Problem 10

Buy 7 cakes from 3 types (combinations with repetition).

$$\binom{7 + 3 - 1}{3 - 1} = \binom{9}{2} = 36$$

Problem 11

Distinct rearrangements of the Armenian word for “mathematics” (10 letters, *matematika* by sound). In Armenian script the two “t” sounds are written with *different* letters, so the only repeats are m ($\times 2$) and a ($\times 3$); the other five letters are all distinct.

$$\frac{10!}{2! 3!} = \frac{3,628,800}{12} = 302,400$$

Problem 12

Arrange 15 math, 16 informatics, 12 physics books on a shelf (books of the same subject are indistinguishable). Total $15 + 16 + 12 = 43$ books.

$$\frac{43!}{15! \cdot 16! \cdot 12!} = \binom{43}{15, 16, 12}$$

Problem 13

Coefficient of $x_1^4 x_2^3 x_3^2$ in $(x_1 + x_2 + x_3)^9$.

By the multinomial theorem: $\frac{9!}{4! 3! 2!}$.

$$\frac{9!}{4! 3! 2!} = \frac{362880}{24 \cdot 6 \cdot 2} = 1260$$

Practice 7: Partitions & Inclusion-Exclusion

Problem 1

Partition a 5-element set into 3 non-empty classes: Stirling number $S(5, 3)$.

$$S(5, 3) = S(4, 2) + 3 \cdot S(4, 3) = 7 + 3 \cdot 6 = \boxed{25}.$$

Problem 2

Distribute m distinct balls into n distinct boxes, no box empty (surjections $m \rightarrow n$).

By inclusion-exclusion:

$$n! S(m, n) = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$$

Problem 3

Numbers in $\{1, \dots, 1000\}$ not divisible by 2, 3, or 5.

Let A_d = multiples of d . By inclusion-exclusion:

$$\begin{aligned} |A_2| &= 500, |A_3| = 333, |A_5| = 200, \\ |A_6| &= 166, |A_{10}| = 100, |A_{15}| = 66, |A_{30}| = 33. \end{aligned}$$

$$|A_2 \cup A_3 \cup A_5| = 500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

$$\boxed{1000 - 734 = 266}$$

Problem 4

4-digit numbers containing at least one digit 7.

Total 4-digit numbers: 9000. Those with no 7: first digit 8 choices, each other digit 9 choices $= 8 \times 9^3 = 5832$.

$$\boxed{9000 - 5832 = 3168}$$

Problem 5

Integers in $\{1, \dots, 16500\}$ not divisible by 5.

Divisible by 5: $\lfloor 16500/5 \rfloor = 3300$.

$$\boxed{16500 - 3300 = 13200}$$

Problem 6

Integers in $\{1, \dots, 16500\}$ divisible by neither 5 nor 3.

$$|A_3| = 5500, |A_5| = 3300, |A_{15}| = 1100. |A_3 \cup A_5| = 5500 + 3300 - 1100 = 7700.$$

$$\boxed{16500 - 7700 = 8800}$$

Problem 7

Integers in $\{1, \dots, 16500\}$: divisible by 11, not by 5, not by 3.

Universe S : multiples of 11, $|S| = 1500$. Let A = also div. by 3 (div. by 33), B = also div. by 5 (div. by 55). $|A| = 500$, $|B| = 300$, $|A \cap B| = \lfloor 16500/165 \rfloor = 100$. $|A \cup B| = 700$.

$$\boxed{1500 - 700 = 800}$$

Problem 8

Integers in $\{1, \dots, 16500\}$ not divisible by 5, 3, or 11.

$|A_3| = 5500$, $|A_5| = 3300$, $|A_{11}| = 1500$. $|A_{15}| = 1100$, $|A_{33}| = 500$, $|A_{55}| = 300$. $|A_{165}| = 100$.

$|A_3 \cup A_5 \cup A_{11}| = 5500 + 3300 + 1500 - 1100 - 500 - 300 + 100 = 8500$.

$$\boxed{16500 - 8500 = 8000}$$

Problem 9

30 students; 20 know English, 12 French, 6 both.

$|E \cup F| = 20 + 12 - 6 = 26$. Know neither:

$$\boxed{30 - 26 = 4}$$

Problem 10

Each side of $\triangle ABC$ is divided into 8 equal parts. Count triangles whose vertices are division points, with A, B, C *not* used as vertices and *no* side parallel to a side of ABC .

Vertices come from the 7 interior division points on each side (21 points in all). If two vertices lay on the same side, that side of the triangle would be parallel to a side of ABC ; so a valid triangle has *exactly one vertex on each side* — $7^3 = 343$ triangles (none degenerate).

Now subtract those with a side parallel to ABC . A triangle's side is parallel to one specific side of ABC in $7 \cdot 7 = 49$ cases (one vertex free, the other two constrained to match). By inclusion–exclusion over the three side-directions ($|A_i| = 49$, $|A_i \cap A_j| = 7$, $|A_1 \cap A_2 \cap A_3| = 1$):

$$343 - (3 \cdot 49 - 3 \cdot 7 + 1) = 343 - 127 = \boxed{216}.$$

Problem 11

Three carpets of area 3 m^2 each in a room of area 6 m^2 . Prove two of them overlap by $\geq 1 \text{ m}^2$.

Proof sketch. Total carpet area = 9 m^2 , room area = 6 m^2 , so the total overlap (counted with multiplicity) is $\geq 9 - 6 = 3 \text{ m}^2$. There are $\binom{3}{2} = 3$ pairs of carpets. By the Pigeonhole Principle, at least one pair overlaps by $\geq 3/3 = 1 \text{ m}^2$. $\square$

Problem 12

Integer solutions of $x_1 + x_2 + x_3 = 40$ with $4 \leq x_1 \leq 15$, $9 \leq x_2 \leq 18$, $5 \leq x_3 \leq 16$.

Substitute $y_i = x_i$ – lower bound: $y_1 + y_2 + y_3 = 22$ with $0 \leq y_1 \leq 11$, $0 \leq y_2 \leq 9$, $0 \leq y_3 \leq 11$.

Unrestricted: $\binom{24}{2} = 276$. Violations by inclusion–exclusion: $|A_1| + |A_2| + |A_3| - (|A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3|) + |A_1 \cap A_2 \cap A_3| = (66 + 91 + 66) - (1 + 0 + 1) + 0 = 221$.

$$\boxed{276 - 221 = 55}$$

Practice 8: Recurrence Relations

Problem 1

$$a_1 = 3, a_2 = 9, a_{n+2} = a_{n+1} - a_n.$$

The sequence is periodic with period 6: 3, 9, 6, -3, -9, -6, 3, 9, ...

Since $200 = 33 \times 6 + 2$, we have $a_{200} = a_2$.

$$a_{200} = 9$$

Problem 2

$$a_n = 5a_{n-1} - 6a_{n-2}, a_1 = 2, a_2 = 10.$$

Characteristic equation: $r^2 - 5r + 6 = 0 \Rightarrow r = 2, 3$. General solution: $a_n = \alpha \cdot 2^n + \beta \cdot 3^n$.

From initial conditions: $\alpha = -2, \beta = 2$.

$$a_n = 2(3^n - 2^n)$$

Problem 3

$$a_n = 4a_{n-1} - 4a_{n-2}, a_0 = 2, a_1 = 6.$$

Characteristic equation: $r^2 - 4r + 4 = 0 \Rightarrow r = 2$ (double root). General solution: $a_n = (\alpha + \beta n) \cdot 2^n$. $a_0 = \alpha = 2, a_1 = (\alpha + \beta) \cdot 2 = 2(2 + \beta) = 6 \Rightarrow \beta = 1$.

$$a_n = (2 + n) \cdot 2^n$$

Problem 4

$$a_n = 2\sqrt{2}a_{n-1} - 4a_{n-2}, a_0 = 1, a_1 = 2.$$

Characteristic equation: $r^2 - 2\sqrt{2}r + 4 = 0$. $r = \sqrt{2}(1 \pm i)$. In polar form: $\rho = 2, \theta = \pi/4$.

$$a_0 = \alpha = 1. a_1 = \sqrt{2}(1 + \beta) = 2 \Rightarrow \beta = \sqrt{2} - 1.$$

$$a_n = 2^n \left(\cos \frac{n\pi}{4} + (\sqrt{2} - 1) \sin \frac{n\pi}{4} \right)$$

Problem 5

$$a_n = 2a_{n-1} + a_{n-2}, a_0 = 0, a_1 = 1.$$

Characteristic equation: $r^2 - 2r - 1 = 0 \Rightarrow r = 1 \pm \sqrt{2}$. $a_0 = \alpha + \beta = 0, a_1 = (1 + \sqrt{2})\alpha + (1 - \sqrt{2})\beta = 2\sqrt{2}\alpha = 1$, so $\alpha = \frac{1}{2\sqrt{2}}$.

$$a_n = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

Problem 6

Binary strings of length 10 containing (at least) two consecutive zeros.

By complementary counting: let b_n = strings of length n with no "00". $b_n = b_{n-1} + b_{n-2}$ (Fibonacci-like), $b_1 = 2, b_2 = 3, b_{10} = 144$. Total strings: $2^{10} = 1024$.

$$1024 - 144 = 880$$

Problem 7

Words of length n from $\{a, b, c\}$ such that a and b are never adjacent.

Let A_n, B_n, C_n = words ending in a, b, c with no adjacent a, b . $A_n = A_{n-1} + C_{n-1}$, $B_n = B_{n-1} + C_{n-1}$, $C_n = A_{n-1} + B_{n-1} + C_{n-1}$. $A_1 = B_1 = C_1 = 1$, so $f(1) = 3$.

The total $f(n) = A_n + B_n + C_n$ satisfies $f(n) = 2f(n-1) + f(n-2)$ with $f(1) = 3, f(2) = 7$.

$$f(n) = \frac{(1 + \sqrt{2})^{n+1} + (1 - \sqrt{2})^{n+1}}{2}$$

(For specific lengths: $f(10) = 8119$.)